Staff answers · Supported

Arrival:1 membranes;2 larger tissue debris;3 an observation;4 trained adult following local risk assessment. Aspire gives a valid reason using model/evidence.

Explain: mashing breaks tissue; detergent disrupts membranes; salt assists extraction/precipitation; filtering removes large debris; cold ethanol helps DNA precipitate. Adult controls the ethanol and all liquids.

Record: supplied record states white strands appeared at the boundary; consistent with a crude DNA-containing extract. Do not rewrite supplied data as the pupil's own observed result.

Evaluate: appearance does not show purity. Keep conditions consistent and use a useful comparison. A water comparison does not chemically identify DNA. Exit2: other cell material may remain; appearance alone does not establish purity.

Exit1: detergent helps disrupt membranes, releasing cell contents including DNA. For all routes, record own reasoning, one valid limit and one next step; do not infer independence from clicks or completion.

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Staff answers · Independent

Arrival:1 membranes;2 larger tissue debris;3 an observation;4 trained adult following local risk assessment. Aspire gives a valid reason using model/evidence.

Explain: mashing breaks tissue; detergent disrupts membranes; salt assists extraction/precipitation; filtering removes large debris; cold ethanol helps DNA precipitate. Adult controls the ethanol and all liquids.

Record: supplied record states white strands appeared at the boundary; consistent with a crude DNA-containing extract. Do not rewrite supplied data as the pupil's own observed result.

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Exit1: detergent helps disrupt membranes, releasing cell contents including DNA. For all routes, record own reasoning, one valid limit and one next step; do not infer independence from clicks or completion.

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Staff answers · Aspire

Arrival:1 membranes;2 larger tissue debris;3 an observation;4 trained adult following local risk assessment. Aspire gives a valid reason using model/evidence.

Explain: mashing breaks tissue; detergent disrupts membranes; salt assists extraction/precipitation; filtering removes large debris; cold ethanol helps DNA precipitate. Adult controls the ethanol and all liquids.

Record: supplied record states white strands appeared at the boundary; consistent with a crude DNA-containing extract. Do not rewrite supplied data as the pupil’s own observed result.

Evaluate/Exit2: different fruit masses, volumes, mixing, timing or contaminants can alter visible material. Keep conditions fair and use an appropriate DNA measurement before comparing true yield.

Exit1: detergent helps disrupt membranes, releasing cell contents including DNA. For all routes, record own reasoning, one valid limit and one next step; do not infer independence from clicks or completion.

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Staff answers · models, cards and evidence

Starter: second option, a crude DNA-containing extract may have formed. First and third overclaim visible genes/purity.

Six sorter placements, in order: release/release/separate/separate/judge/judge. Each exact card clue is on the pupil clue page. All six are distinct steps in the method/evidence reasoning.

Four worked steps: clear is appearance; filtering removed large debris, not all dissolved material; invisible other molecules may remain; clear filtrate does not prove pure DNA.

Fictional table: fruit extract white strands at boundary; water no visible strands; no chemical purity test. Water comparison counters strands in every tube; it does not identify all substances or establish true DNA yield.

Reference cards: white strands=direct observation; DNA-containing material=scientific inference based on method knowledge; purity unknown=method limitation. Preserve the distinction and pupil reason.

DNA recap: chromosomes contain long DNA molecules; genes are sections of DNA. Extracted material came from fruit-cell chromosomes and contains many genes. This method does not show individual genes, base sequence/base pairs or naked-eye double helix.

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Staff guidance · method, evidence and identity

Teacher/technician chooses physical adult demonstration, pupil-directed adult handling or supplied-observation route. Label the fruit-DNA activity non-core. Approve local method, chemical concentrations/quantities, PPE and disposal. Adult prepares fruit/solutions and handles all liquids; no pupil direct contact. Do not add quantities or procedural steps from this resource.

Local chemical risk assessment applies. Ethanol is flammable: adult controlled, away from flames/sparks/heat. Check allergy privately; no tasting, mouth pipetting or human samples. Sealed soft-fruit bag/tray; wash hands; staff clear spills and dispose as approved.

Secure→fade content prompts/transfer a limit; mixed→point to deciding evidence and invite revision; misconception→contrast observation with purity claim and recheck; access→change format/mode/adult role while keeping goal; method/evidence→keep gap visible, name approved check/source needed.

Current main DOM+lesson-data stage clock4/3/4/4/3/4/14/4 governs40min. Legacy config.timings3minArrival/15minIndependent and alternate I-can objective are stale. Body/main wrongly W8L1/Introduce; corrected to W12L3/Do, taught Autumn2Week4. Stale amylase pH/iodine print residue and W8L2next-link excluded by recorded correction.

Existing six W12L3 hub members examined: old17slide deck and teacher/pupil notes are not current-DOM-equivalent. Reissue/restructure source content, metadata/credit and standard member routes in CHECKS. Workbook LAUNCH Weekly - Autumn B42Aut2·W4 C42 exact: Describe DNA, genes & chromosomes.

Source Launch/W8-W13_2026-27/SCI_L_W12L3_Fruit_DNA_Evidence_Do.html @ c4ca8ffab98e6186a9ef2fed3921644b177f0d6d. Original source unchanged. No awarding/qualification external link reproduced.

Made from the web lesson SCI-LAUNCH-A2-W04-L3
Version _____ 5

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